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Studierichting: Informatica

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$$\begin{aligned} \int \frac{4x^2 + 2x + 1}{(2x - 1)^3} dx &\stackrel{t=2x-1}{=} \frac{1}{2} \int \frac{4 \left(\frac{1}{2}t + \frac{1}{2} \right)^2 + 2 \left(\frac{1}{2}t + \frac{1}{2} \right) + 1}{t^3} dt \\ &= \frac{1}{2} \int \frac{t^2 + 3t + 3}{t^3} dt = \frac{1}{2} \int \frac{1}{t} dt + \frac{3}{2} \int \frac{1}{t^2} dt + \frac{3}{2} \int \frac{1}{t^3} dt \\ &= \frac{1}{2} \ln |t| - \frac{3}{2t} - \frac{3}{4t^2} + c \stackrel{t=2x-1}{=} \frac{1}{2} \ln |2x - 1| - \frac{3}{2(2x - 1)} - \frac{3}{4(2x - 1)^2} + c \end{aligned}$$

References